

CSE446: Blockchain & Cryptocurrencies

Lecture – 11: Other Consensus Algorithms



Inspiring Excellence

Agenda

- PoW Limitations
- Other consensus algorithms
 - Proof of Stake (PoS)
 - Delegated Proof of Stake (DPoS)

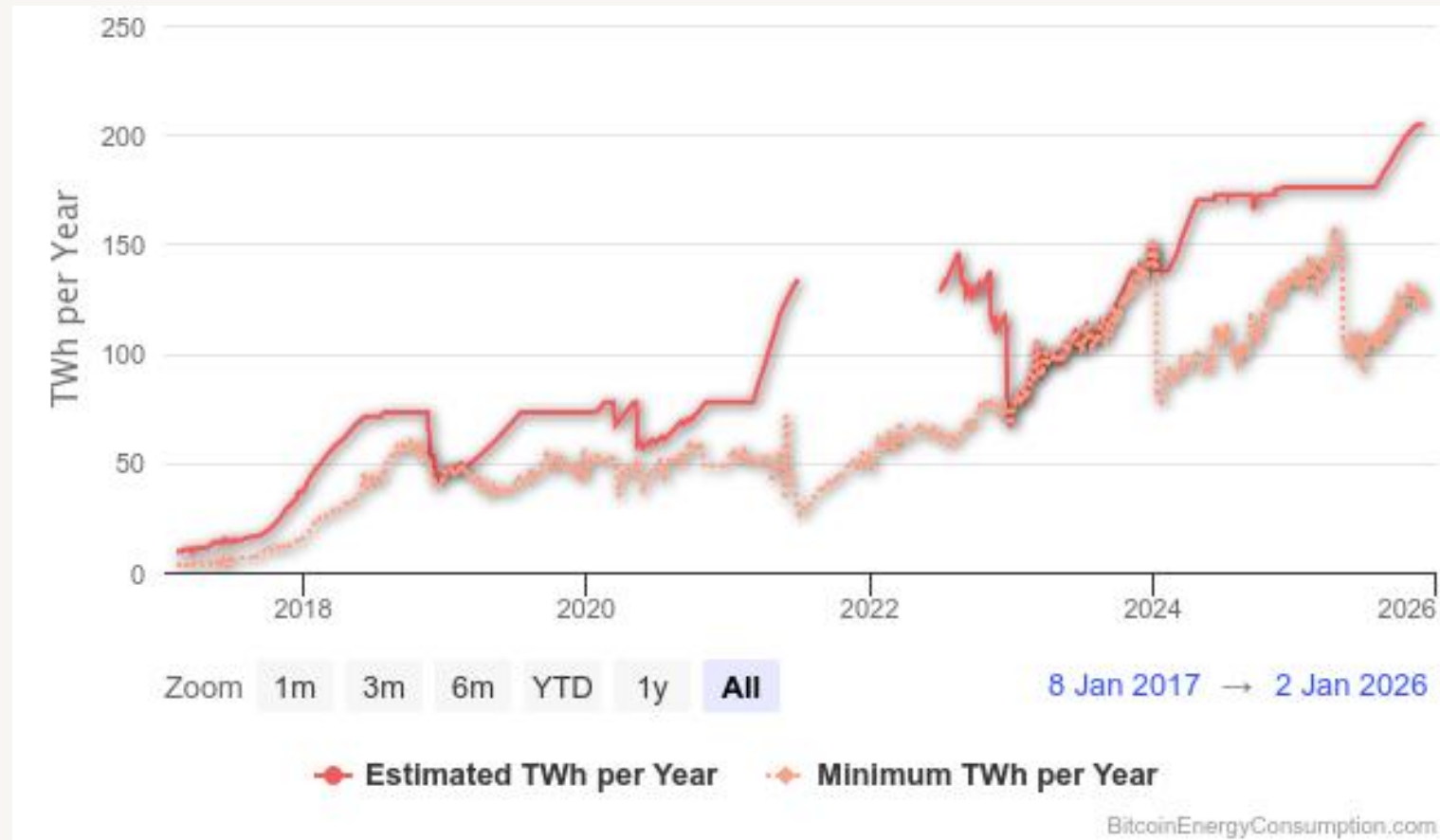
PoW Limitations

- There are a few major limitations of PoW
 - Energy consumption
 - Absence of penalty
 - Delay in block finality (confirmation)

PoW Limitations: energy consumption

- Each PoW algorithm needs to consume electricity to compute the hash
- As the difficulty of the network starts to increase, so does the energy consumption
- The amount of consumed energy is quite significant when calculated over the whole network consisting of ASIC/GPU mining rigs all around the world

PoW Limitations: energy consumption



PoW Limitations: energy consumption

Single Bitcoin Transaction Footprints

Carbon Footprint

534.77 kgCO₂



Equivalent to the carbon footprint of 1,185,234 VISA transactions or 89,128 hours of watching Youtube.

Electrical Energy

958.78 kWh



Equivalent to the power consumption of an average U.S. household over 32.86 days.

Electronic Waste

219.90 grams



Equivalent to the weight of 1.34 iPhones 12 or 0.45 iPads. (Find more info on e-waste [here](#).)

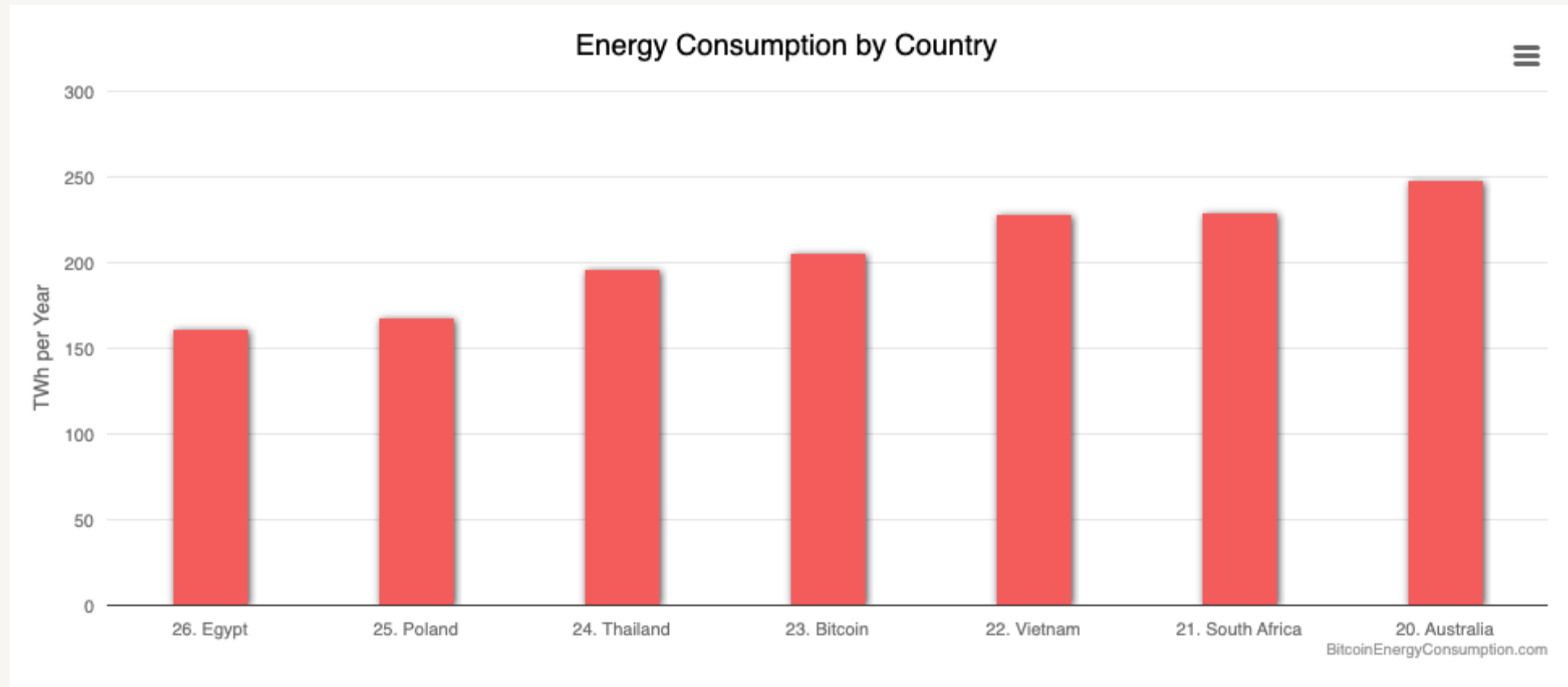
Fresh Water Consumption

15,110 liters



Equivalent to the amount of water in a backyard swimming pool.

PoW Limitations: energy consumption by country



PoW Limitations: energy consumption by country



J64 ✓ @_J64_ · Dec 1



“Energy is the true currency. This is why I said **Bitcoin** is based on energy.
[@elonmusk](#)

[@nikhilkamathcio](#) discussion with **Elon** that highlights the paradigm shift of how energy is the purest form of value. **Bitcoin** is the future. **Elon** knows it.

PoW Limitations: absence of penalty

- PoW algorithms are altruistic in nature in the sense that they reward behaving miners
 - However, they do not penalise a misbehaving miner
- One example is that a miner can collude with a group of miners (a phenomenon known as selfish mining) to increase its profitability in an illegitimate way

PoW Limitations: absence of penalty

- In addition, a miner can engage in Denial-of-Service attacks by just not forwarding any transaction or block within the network
- Furthermore, such malicious miners can join forces to engage in the spawn-camping attack
 - Launching DoS attacks simultaneously over and over again to render the blockchain network useless
- A penalty mechanism would disincentivise any miner to engage in any type of malicious misbehaviour

PoW Limitations: delay in finality

- Finality is the assurance or guarantee that crypto-currency transactions cannot be altered, reversed, or cancelled after they are completed
- Finality is used to measure the amount of time one has to wait for a reasonable guarantee for a transaction to be confirmed (included in a block)
- In blockchain technology, transactions are termed immutable due to its finality nature
- The latency level of a blockchain will ultimately affect the chain's finality rate

PoW Limitations: delay in finality

- Finality is an essential feature for ventures accepting cryptocurrencies because
 - waiting endlessly on a blockchain network can have a high adverse effect for businesses or enterprises that accept crypto as a means of payment
- When creating a payment system, to be effective, it is crucial to have low latency
- If you were to have to wait for 10 minutes every time you wished to purchase anything, it would quickly become very inconvenient to go shopping
- However, most PoW-based blockchain protocols only show a probabilistic transaction finality
 - meaning that transactions are not automatically or instantly final but become "more and more final" over time (as more blocks are confirmed)
 - For Bitcoin, it is estimated that one has to wait 6 blocks, around 1 hour, before we can say that a transaction is final with a reasonable guarantee

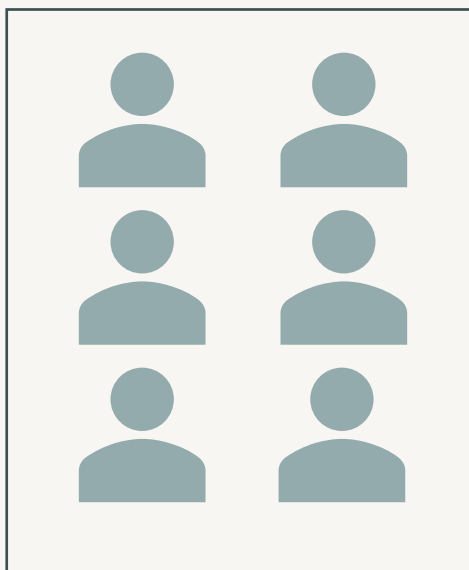
Proof of Stake (PoS)

- In PoS, the nodes who would like to participate in the block creation process must prove that they own a certain number of coins at first
- Besides, they must lock a certain amount of its currencies, called stake, into an escrow account in order to participate in the block creation process
- The stake acts as a guarantee that it will behave as per the protocol rules
- The node that escrows its stake in this manner is known as the stakeholder, staker, validator, leader, forger, or minter in PoS terminology
- The minter can lose the stake, in case it misbehaves

Proof of Stake

- In essence, when a stakeholder escrows its stake, it implicitly becomes a member of an exclusive group
- Only a member of this exclusive group can participate in the block creation process
- How much block a minter can generate depends on their size of stakes
- The stakeholder who produces blocks are rewarded in one of the two different ways
 - Either it can collect the transaction fees within the block, or
 - It is provided a certain amount of currencies that act as a type of interest against their stake

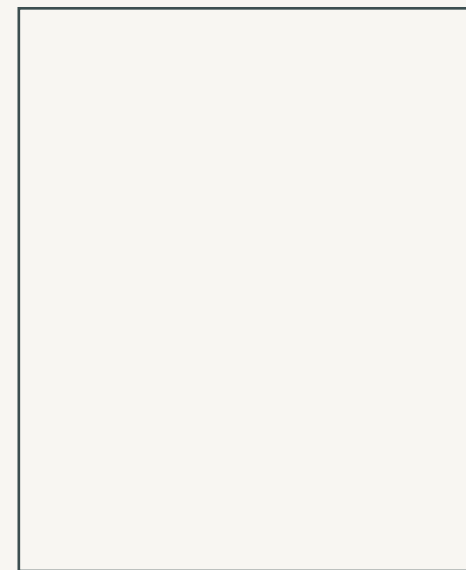
Proof of Stake



Set of Users

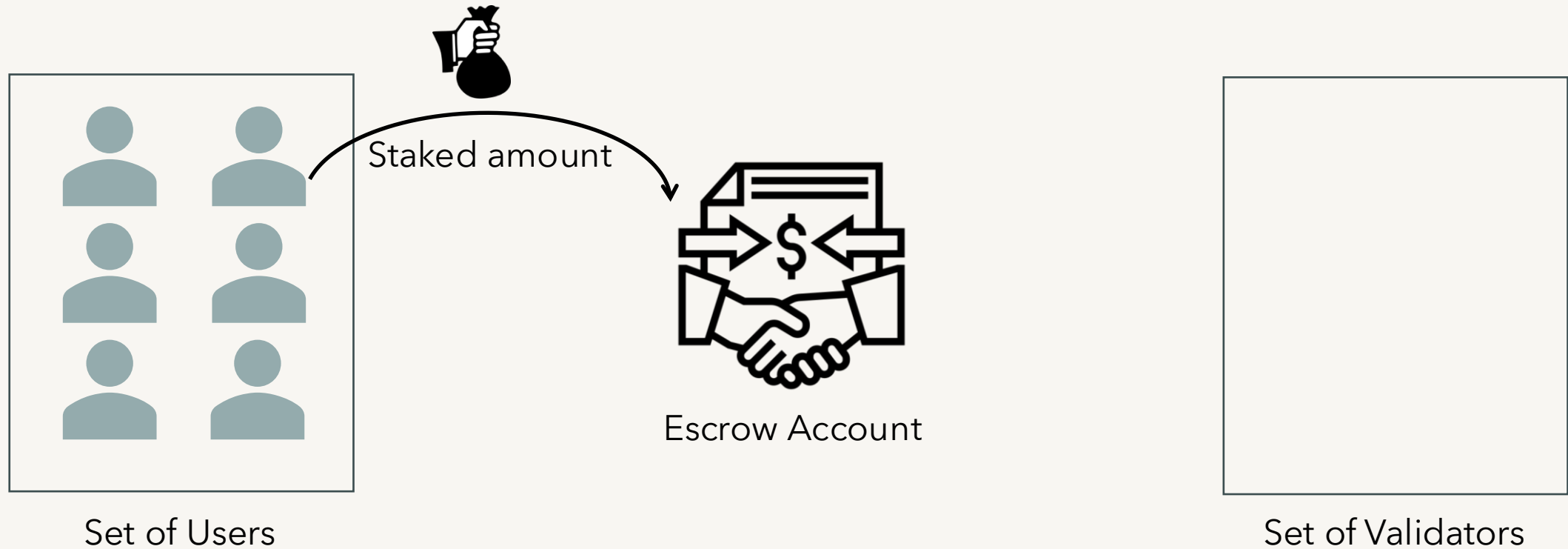


Escrow Account

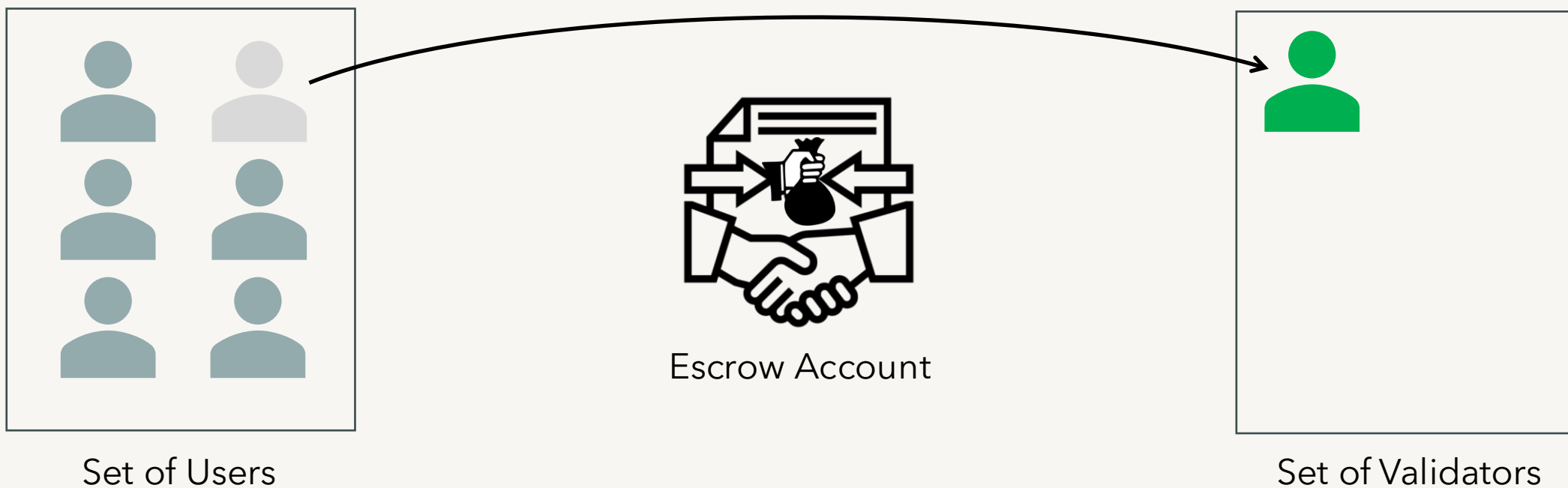


Set of Validators

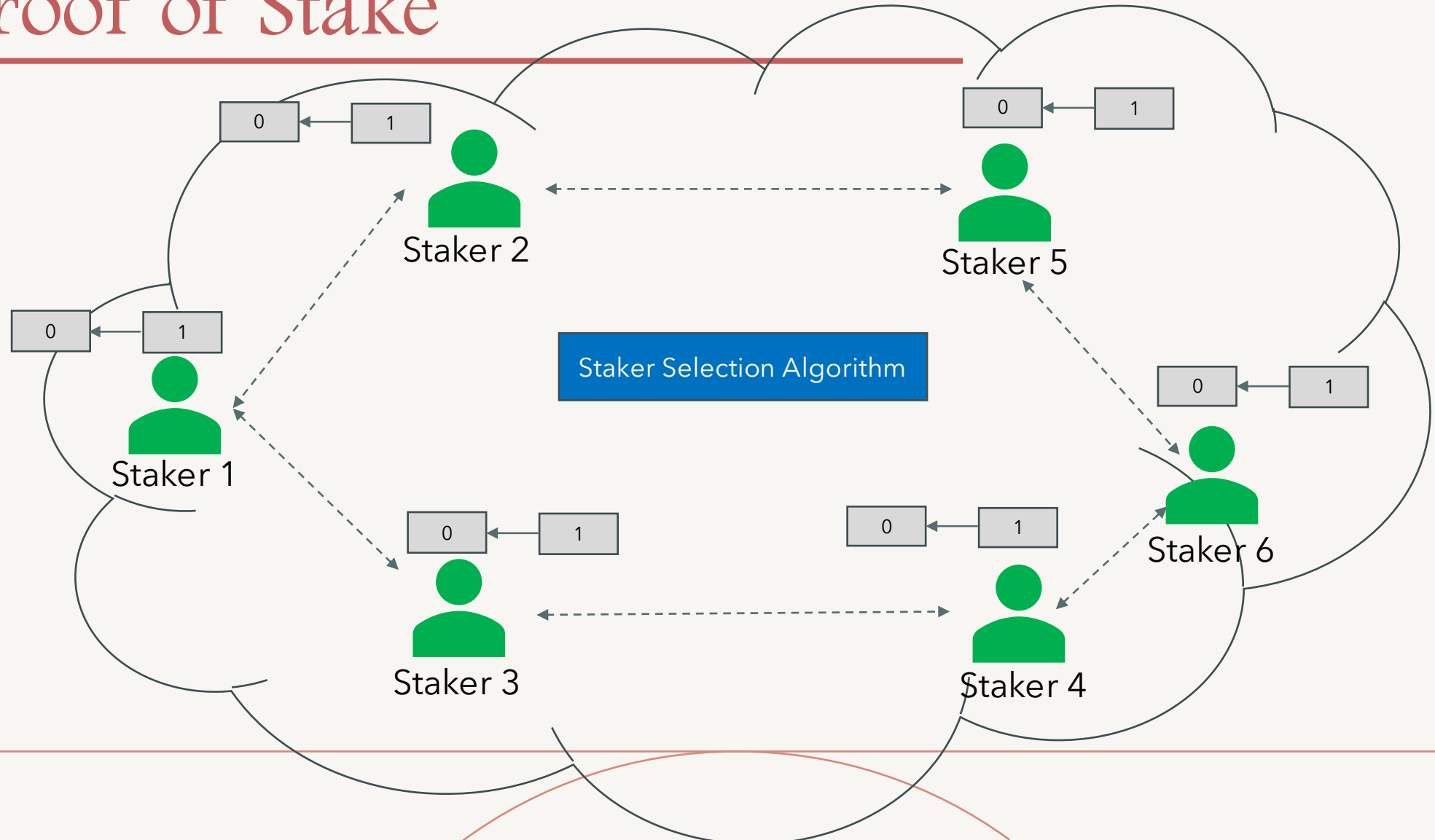
Proof of Stake



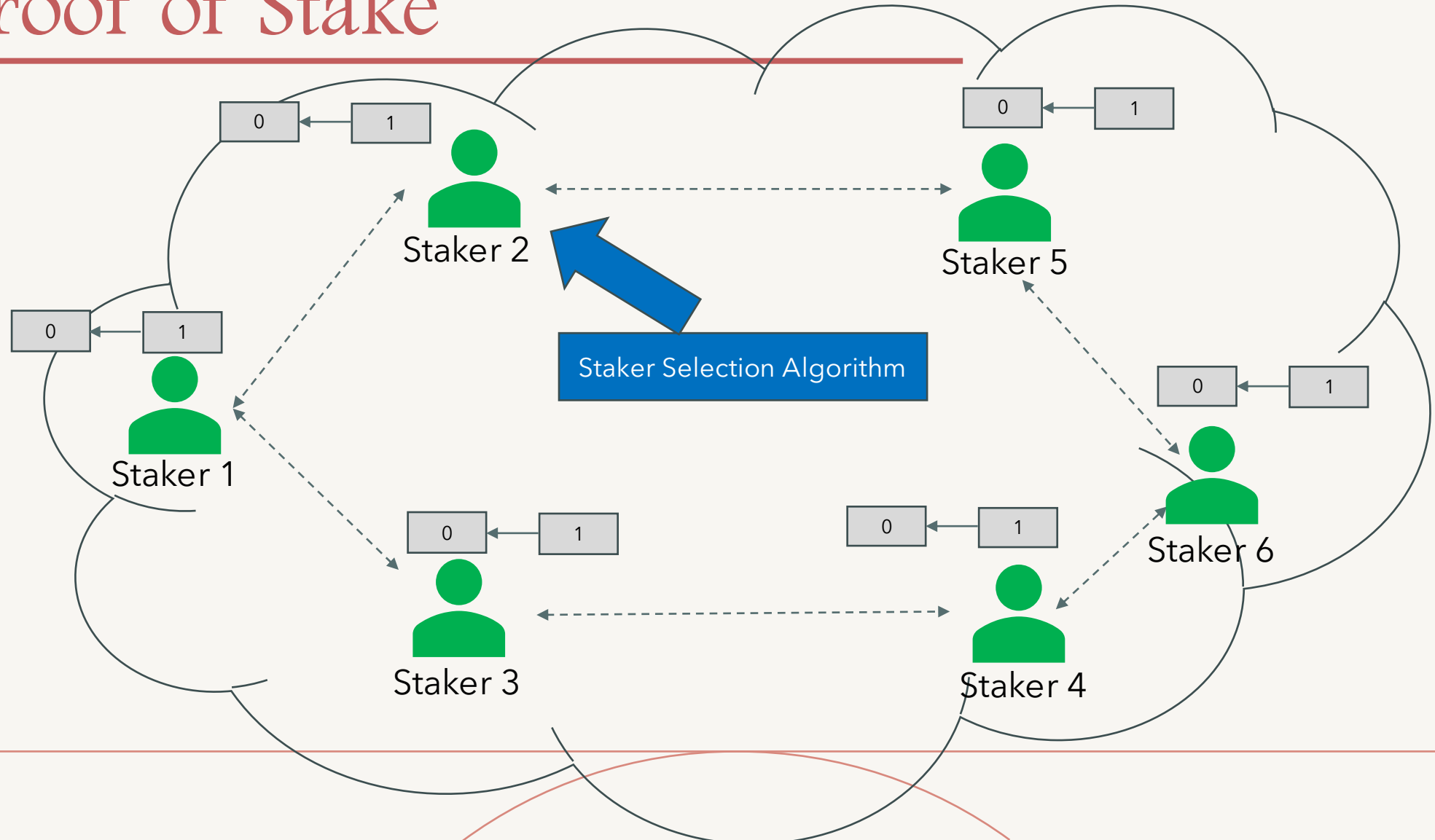
Proof of Stake



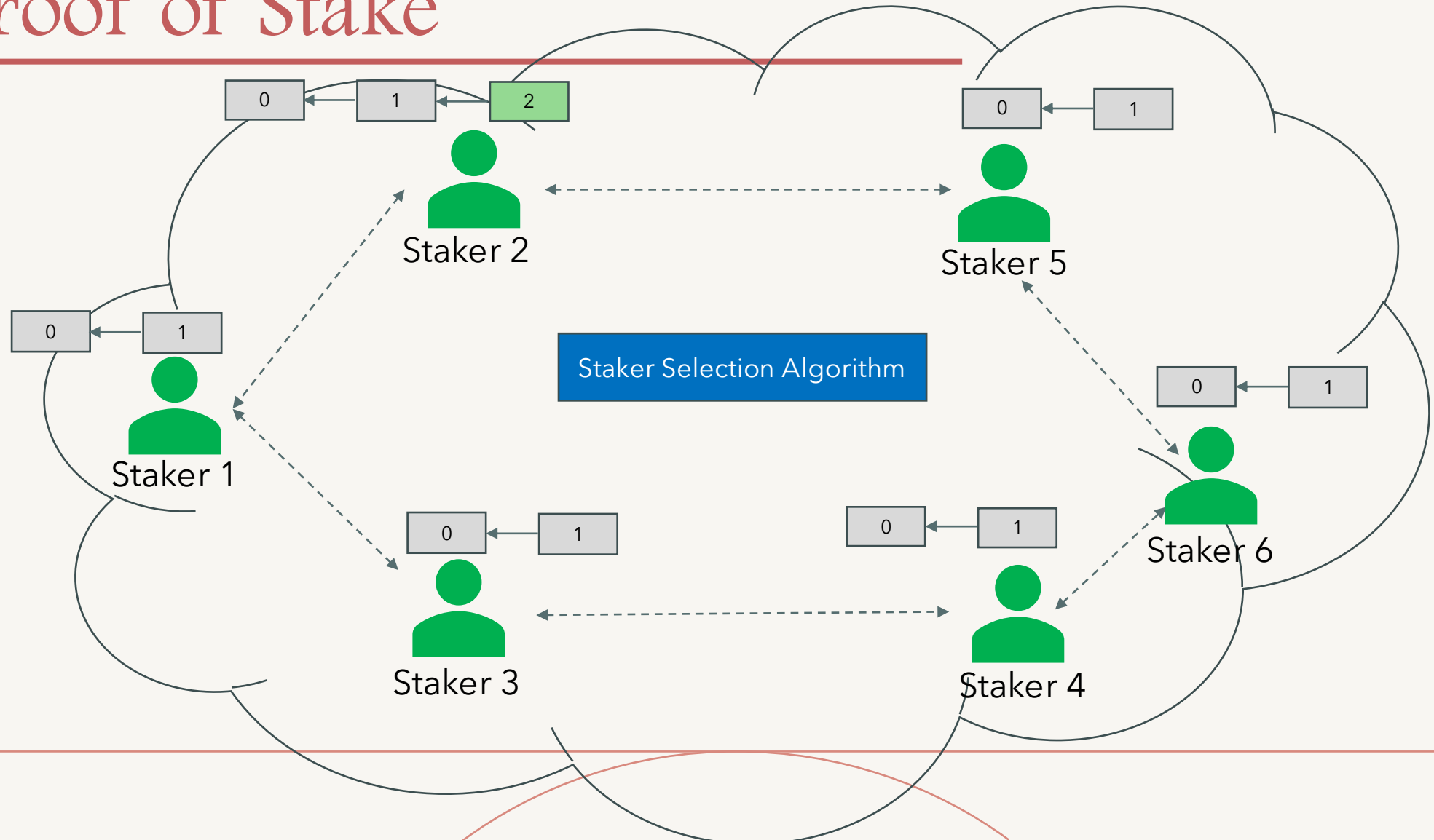
Proof of Stake



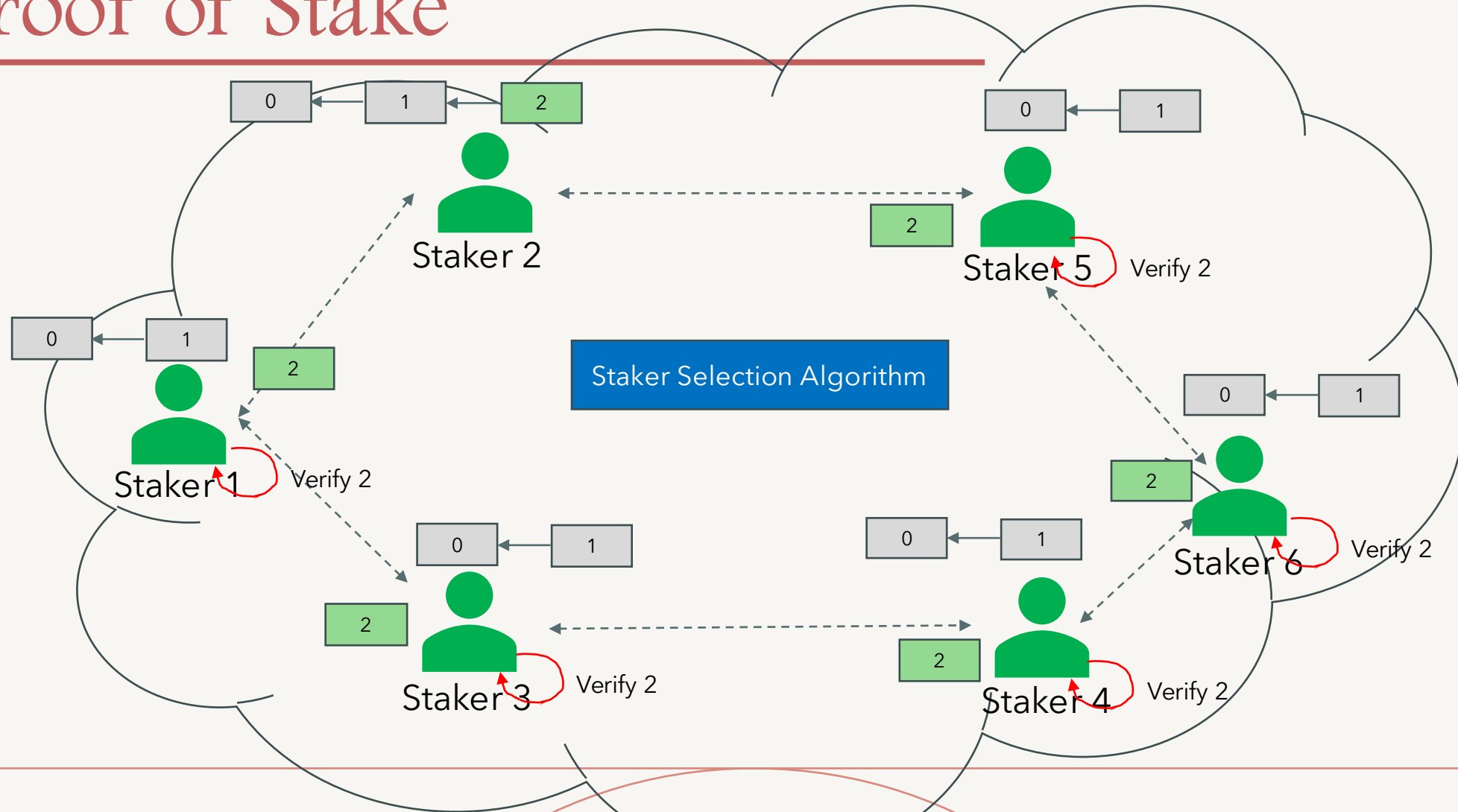
Proof of Stake



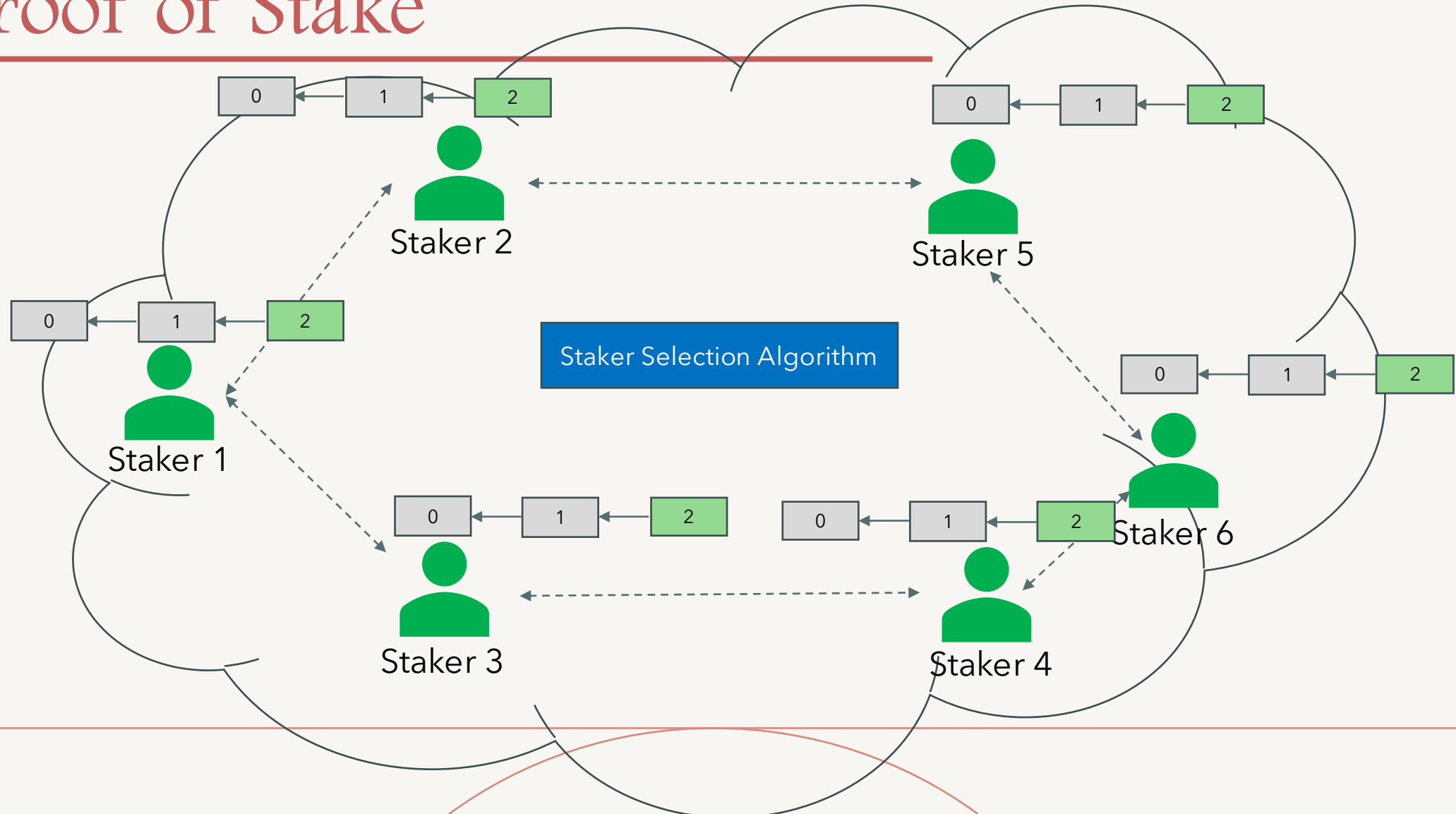
Proof of Stake



Proof of Stake



Proof of Stake



PoS: Bootstrapping issue

- One of the major barriers in a PoS algorithm is how to generate the initial coins
- A fair distribution of coins among the stakeholders are essential to ensure a secure PoS algorithm
- This is known as the bootstrapping problem
- There are two ways to solve the bootstrap issue:
 - Pre-mining
 - PoW to PoS transition

PoS: Bootstrapping issue

- Pre-mining:
 - A set of coins are pre-mined, which are then sold before the launch of the system in an IPO (Initial Public Offering) or ICO (Initial Coin Offering)
- PoW-PoS transition
 - The system starts with a PoW system to fairly distribute the coins among the stakeholders
 - Then, it slowly transitions towards the PoS system
 - Ethereum took this approach

DPoS

- Delegated Proof of Stake (or DPoS in short) is a form of consensus algorithm in which reputation scores or other mechanisms are used to select the set of (delegated) validators
- Even though it has the name Proof of Stake associated with it, it is quite different from other PoS algorithms
- In DPoS, users of the network vote to select a group of delegates (or witnesses) who are responsible for creating blocks
- Delegates are the only entities who can propose new blocks

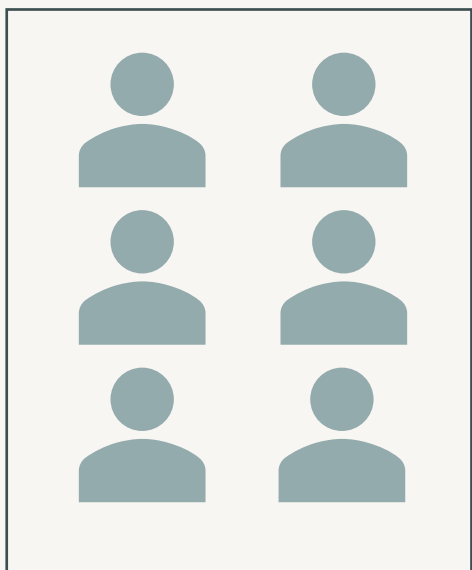
DPoS

- For each round, a leader is selected from the set of delegates who can propose a block
- How such a leader is chosen depends on the respective blockchain system
- The leader gets rewards for creating a new block, and is penalised and de-listed from the set of validators if it misbehaves

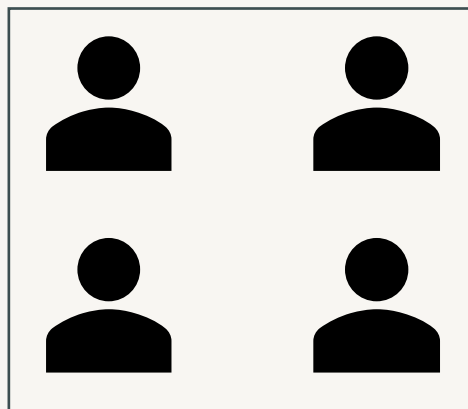
DPoS

- The delegates themselves compete with each other to get included in the validator list
- In such, each validator might offer different levels of incentives for the voters who vote for it
- For example, if a delegate is selected to propose a block, it might distribute a certain fraction of its reward among the users who have voted for the delegate
- Since the number of validators is small, the consensus finality (confirmation) can be fast

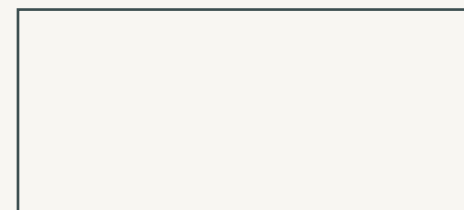
DPoS



Set of Users

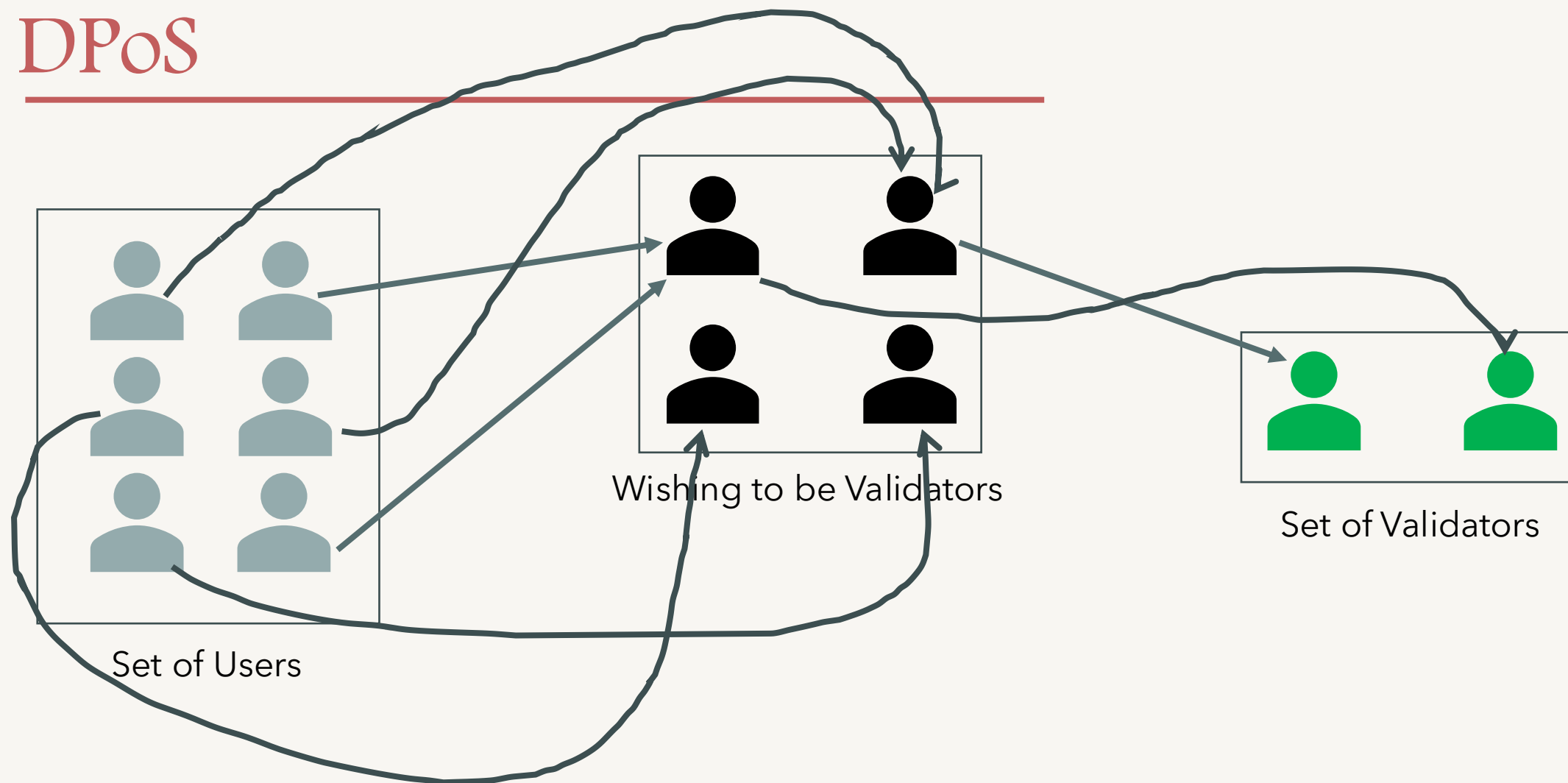


Wishing to be Validators



Set of Validators

DPoS



DPoS: EOS

- EOS is the first DPoS crypto-currency and smart-contract platform
- The DPoS consensus algorithm of EOS utilises 21 validators, also known as Block Producers (BPs)
- These 21 validators are selected with votes from EOS token (currency) holders
- The number of times a particular BP (block producer) is selected to produce a block is proportional to the total votes received from the token holders

DPoS: EOS

- Blocks in EOS are produced in rounds where each round consists of 21 blocks
- At the beginning of each round, 21 BPs are selected
- Next, each of them gets a chance to create a block in pseudo-random fashion within that particular round
- Once a BP produces a block, other BPs must validate the block and reach into a consensus
- A block is confirmed only when the (+2/3) majority of the BPs reach the consensus regarding the validity of the block
- Once this happens, the block and the associated transactions are regarded as confirmed or final, so no fork can happen

Finality: Bitcoin vs EOS

Blockchain	Consensus	Avg. time per block	Avg. time to finality
Bitcoin	PoW	10 minutes	60 minutes (6 confirmations)
EOS	DPoS	0.5 - 1 second	2-3 seconds (2-3 commitments)

ANY QUESTIONS?

